

# Master Thesis Proposal

Topic: Analyzing the career length of professional soccer players by means of survival analysis

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## 1. Introduction

Survival Statistics, also known as time-to-event analysis, is a branch of statistics that focuses on the time until an event of interest occurs. This method is effective at handling censoring, a key analytical challenge when some subjects' survival times are unknown [13]. Although this method was popularized in medical studies, it is now applied across different fields.

In this thesis project, we will analyze the career duration of professional football players in two major European leagues, accounting for right-censored data as a portion of the players will still be active at the time of the data collection. The influence of different covariates on the length of the careers will be considered, including demographic factors, career-related factors such as position in the field, debut age and also injury history.

Player data will be collected through web scraping techniques from two independent public sources; the sources will be cross-checked and validated to ensure accuracy and robustness. Then, statistical and machine learning models will be applied with a focus on survival analysis techniques.

### 1.1 Football Context

Even though Survival statistics' roots were in clinical trials and reliability, its flexibility has led to application in a variety of fields, such as economics, sociology and more recently in sports analytics as well.

In that last field, researchers have used this approach in different sports. For instance, in the analysis of career length of NBA basketball players based on their initial draft number [14] or studying the influence of covariates such as biological and performance factors. [11]

In terms of football, survival analysis has also been used to review career length, considering injury history as covariate and analysing the effect on performance and longevity on players who played three or more games in the top four English football leagues (Premier League, Championship, League One, and League Two). [12]

Football is the most widely followed sport in the world, exhibiting a highly developed professional structure and global fanbase. The 2022 FIFA World Cup engaged 5 billion viewers being the most-watched sporting event in the world, followed by the Tour de France with 3,5 billion viewers.[10]

European Football dominates the football economy. In the 2023-24 season, clubs in the ‘Big Five’ European leagues generated 20.4 billion in revenue, representing 53% of the total European football market. The term “Big Five” refers to the football leagues of England, France, Germany, Italy and Spain. They are considered the European Leaders due to their size, global popularity, financial strength and competitive level of their main domestic football leagues. [9]

Commercial revenue is the primary source of revenue in football, but in 2023-24, profit in player trading presented a significant increase in Europe [9]. Players represent the main assets for clubs, being the main drivers of sporting performance and financial value. Consequently, their career development and longevity are critical topics in sports management.

## 1.2 Research Questions/Objectives

1. *Which player- and career-related characteristics are associated with the overall duration of a professional soccer career and the instantaneous risk of retirement?*

This question focuses on baseline and relatively stable covariates like age at debut, playing position, league level, nationality, long-run performance indicators etc., and how they relate to career length and retirement hazard.

2. *To what extent do the determinants of career length differ between the two leagues under study and across key player subgroups?*

In particular, how do average and/or median career durations, as well as the main risk factors for retirement, vary by league, playing position, and wherever relevant, geographic background?

3. *How does the risk profile of a professional soccer career evolve over time, and how do major time-dependent events affect the probability of continuation and the hazard of retirement?*

For example changes in survival probabilities and retirement risk with age and experience, and the impact of specific events such as serious injuries, surgeries, transfers, or disciplinary sanctions.

4. *How do methodological choices, especially the handling of censored observations and the specification and validation of survival models, influence the estimated career durations and the substantive conclusions?*

Justifying covariate selection using prior literature and exploratory data analysis, deciding how to treat players who remain active or are lost to follow-up, and assessing model adequacy through diagnostic checks and sensitivity analyses (e.g. tests of the proportional hazards assumption and alternative model specifications).

## 2. Literature Review and Methodological Overview

### 2.1 Survival Statistics and Key Concepts

Survival analysis is a branch of statistics for modelling time-to-event data, explicitly accounting for censored observations such as players who are still active or lost to follow-up at the end of the study [13][17]. Core concepts include the *survival function*, which gives the probability that a

player remains active beyond a given time, and the *hazard function*, which describes the instantaneous risk of career termination at a given time conditional on still being active [13]. Censoring arises when the event of interest, in this case career termination, is not fully observed, which is common in football due to ongoing careers or incomplete records. In addition, many relevant variables change over the course of a player's career like age, league level, injury status, so the use of time-dependent covariates and extended survival models is often appropriate [8].

## 2.2 Statistical Methods

The analysis relies on three main methodological components.

### Kaplan–Meier estimator

The Kaplan–Meier estimator is a non-parametric estimator of the survival function, widely used to summarize “still playing” probabilities over time and to compare groups such as positions or leagues [7][13]. In this thesis, it is primarily employed for exploratory analysis and visual comparison, as it does not adjust for multiple covariates simultaneously.

### Cox proportional hazards model

The Cox proportional hazards model is a semi-parametric regression framework in which the hazard at time  $t$  for a player with covariate vector  $X$  is written as:

$$h(t | X) = h_0(t) \cdot \exp(\beta^T X)$$

where  $h_0(t)$  is the baseline hazard and  $\beta$  quantifies the effects of covariates [6][17]. This model allows the simultaneous assessment of several predictors, yields interpretable hazard ratios, and supports formal hypothesis testing for individual covariates. The proportional hazards assumption will be evaluated using diagnostic plots and statistical tests as recommended in the applied survival literature [13].

### Selecting Model Covariates

Professional football represents one of the most competitive and physically demanding career paths in sports. With an average career duration of 5.2 to 11 years, significantly shorter than most professions.[3] Understanding the determinants of career duration is therefore crucial for clubs, as premature retirement can significantly disrupt contract planning, squad development and long-term strategic objectives. [1][2].

The landscape of professional football careers is complex. While elite players in top leagues like the English Premier League enjoy average career lengths of 14.8 years if they also represent their national team, players in lower leagues face much steeper challenges, averaging just 6.2 years [3]. Additionally, career length varies substantially by position. Studies of elite European competitions show that goalkeepers and central defenders tend to be older on average and reach peak performance later than attacking players, while wingers and full-backs are typically

younger and experience earlier performance decline, reflecting the different physical and tactical demands of these roles [5][4].

Beyond age and position, numerous factors shape career trajectories. Severe injuries such as anterior cruciate ligament (ACL) ruptures are among the most serious, typically resulting in many months of time-loss and, in some cases, reduced performance after return to play[15][16]. In addition to including health-related and cumulative career variables in the dataset, covariate selection in the multivariable survival models will be guided by information-criterion measures such as the Akaike Information Criterion (AIC), balancing model fit and parsimony when deciding which predictors to retain.

Thus health related variables and career sum variables were added to the dataset

## 3. Current Progress

### 3.1 Data Collection and Scraping

This research draws on detailed career and match data for professional footballers and the data collection process was designed to ensure reproducibility

#### Data Sources:

- Transfermarkt: Used for gathering comprehensive player career histories, including club transfers, appearances, positions, and reported injuries or suspensions.
- FBref: Utilized for obtaining in-depth player match statistics (minutes played, goals, assists), league participation, and additional performance indicators.

Combined, these sources provide a rich, longitudinal view necessary for robust survival analysis.

#### Scraping Methodology and Technical Steps:

- All data collection scripts and relevant code are available in the project's public repositories ([MichelleHernandez16/KUL\\_Thesis](#) and [GarimaSharda/Thesis](#)), supporting transparency and reproducibility of the research process. These repositories include exploratory notebooks, scraping routines, and data cleaning workflows, allowing other researchers to audit, replicate, or extend the study's findings.
- Tools utilized include:
  - Python (BeautifulSoup and Selenium): For parsing HTML content and Javascript from Transfermarkt, handling pagination, and systematically extracting structured player data.
  - R (rvest and httr): For collecting tabular statistics and player metadata from FBref, including HTTP requests and web-page parsing.

#### Reproducibility and Future Updates:

- The repository README includes usage instructions and setup guides for rerunning the scraping and cleaning pipelines.
- The methods ensure that any interested researcher can replicate data collection or update the dataset as new player seasons are played.

### 3.1.1 Challenges in Webscraping

Web scraping soccer data presents several technical and practical challenges:

- **Dynamic content and JavaScript rendering:** Many soccer statistics websites load data dynamically using JavaScript, requiring browser automation tools rather than simple HTTP requests.
- **Rate limiting and access controls:** Websites often implement rate-limiting to prevent automated scraping. Managing request frequency is essential. We were blocked from transfermarkt and fbref multiple times and had to jump IPs and devices to mitigate this problem.
- **Missing or incomplete records:** Particularly for historical players, data may be sparse or incomplete, creating problems for dealing with NA data.
- **Website structure changes:** Periodic updates to website layouts can break existing scraping scripts, requiring ongoing maintenance. For example, during data scraping a script that was working for scraping fbref data has to be periodically updated just to keep things running smoothly

To mitigate these issues, the scraping pipeline will include error handling, logging, and validation checks to identify and document data quality issues.

### 3.1.2 Dataset

In order to address the research questions, two datasets will be analyzed, covering the period from the 2010-11 to the 2014-15 season. This time window was selected to ensure a sufficiently large sample of players, including players who have already retired. As mentioned earlier, the analysis focuses on the English Premier League and La Liga in Spain. A list of all players who participated in at least one season in either league during the timeframe was collected.

Given the huge amount of information available, only the variables we were able to prioritize based on literature review and intuition were kept, particularly variables related to demographics, performance and injuries. The variables are shown on the table below, and were selected for further analysis.

| Variable name | Explanation        | Variable name | Explanation  |
|---------------|--------------------|---------------|--------------|
| player_name   | Name of the player | nation        | Home country |

|                          |                                                      |                                  |                                                   |
|--------------------------|------------------------------------------------------|----------------------------------|---------------------------------------------------|
| height_tra               | Height (cm)                                          | position_tra                     | Playing position (GK, Defender, Midfield, Attack) |
| foot_tra                 | Preferred foot                                       | current_club_tra                 | Current club                                      |
| joined_tra               | Date joined current club                             | contract_expires_tra             | Contract end date                                 |
| debut_age_transfermarket | Age at professional debut (1st or 2nd tier)          | injury_count_total_tra           | Total number of injuries                          |
| days_missed_total_tra    | Total days missed due to injuries                    | games_missed_total_tra           | Total games missed due to injuries                |
| injury_seasons_tra       | Seasons with at least one injury                     | Deceased_tra                     | Deceased status (1 = dead, 0 = alive)             |
| death_date               | Date of death                                        | death_age                        | Age at death                                      |
| retired_tra              | Retirement status (1 = retired, 0 = active/censored) | retirement_date_tra              | Date of retirement                                |
| year_retirement          | Year of retirement                                   | career_lenght_w_actives_and_dead | Career length up to reference date (years)        |
| National_team            | Ever played for national team (1/0)                  | total_seasons_national           | Seasons with national-team appearances            |

|                       |                                               |                              |                                               |
|-----------------------|-----------------------------------------------|------------------------------|-----------------------------------------------|
| matches_national      | National-team matches played                  | minutes_national             | National-team minutes played                  |
| goals_national        | National-team goals scored                    | CAREER Sum of matches_played | Total club matches (domestic + international) |
| CAREER Sum of minutes | Total club minutes (domestic + international) | CAREER Sum of goals          | Total club goals (domestic + international)   |

## 3.2 Exploratory Data Analysis (EDA)

A preliminary exploratory analysis was performed on the datasets in order to evaluate the data quality and understand the distribution of key covariates. For the Premier League, the dataset includes 1,368 players, of whom 57.5% of them are already retired. For La Liga, the dataset includes 1,263 with 57.8% of them already retired, showing a similar pattern.

The mean observed career length is 14.8 seasons with variations across positions and leagues. Injury history is available for approximately 90% of the players and players on average have experienced 8 injuries and missed 331 days due to them. Initial visualization indicates a positive relationship between the number of injuries and the length of the careers.

The age of debut is on average 19.7 with differences across positions, a preliminary inspection suggests a negative relationship between the debut age and the length of the career.

Other covariates that appear to have a significant effect on the career's length is if the player is ambidextrous, and if he plays with the national team of his respective country, as players that have national team appearances tend to have two to four additional playing years.

All these early insights support the feasibility of our posterior survival analysis and further investigation.

## 4. Future Steps

The remainder of the thesis will be organized in clearly defined phases to ensure systematic progress and timely completion:

- Literature review: Deepen the review on career longevity in professional football, survival models, and sports analytics to refine questions and context.

- **Methodology:** Finalize covariates (e.g. position, age, injuries, league level) and, if needed, adopt time-dependent covariates, competing-risks models, and stronger validation methods.
- **Analysis:** Finish cleaning and merging Transfermarkt and FBref data, fit the survival models (Kaplan–Meier, Cox, etc.), run diagnostics, and generate survival and hazard plots.
- **Writing:** Draft and iteratively revise methodology, literature review, results, and discussion, integrating supervisor feedback.
- **Completion:** Prepare the defense presentation, revise the full thesis based on committee input, and finalize submission with complete references and ethics documentation.

| Period   | Main activities                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| December | Finalize literature review on survival analysis and soccer careers; complete initial data cleaning and integration from Transfermarkt and FBref.  |
| January  | Fix list of covariates; refine methodology (model choices, handling of censoring and time-dependent covariates); set up analysis scripts.         |
| February | Fit main survival models (Kaplan–Meier, Cox models, possible extensions); run diagnostics and robustness checks; generate key figures and tables. |
| March    | Interpret results; draft Results and Discussion sections; revise Introduction and Literature Review based on findings and supervisor feedback.    |
| April    | Refine all chapters; add limitations and future work; prepare slides and speaking notes for the defense.                                          |
| May      | Final proofreading and formatting; check references; submit thesis and complete defense requirements.                                             |

## References

[1] FIFPro (2024). Professional Footballer Career Span Study. Retrieved from <https://fifpro.org>



- [2] Professional Footballers' Association (PFA). Career Statistics and Injury Data.  
<https://www.givemesport.com/>
- [3] Jones, A., et al. (2025). Career Length of Professional Male Soccer Players: Comparative Analysis Across League Levels. *Journal of Sports Medicine and Physical Fitness*.
- [4] ScienceDirect & Frontiers in Sports. (2024). Position-Specific Career Trajectories in Professional Soccer.
- [5] Kalén, A., et al. (2019). Aging Trends in Professional Soccer: A Longitudinal Analysis. *British Journal of Sports Medicine*, 48(7), 630–636.
- [6] Cox, D. R. (1972). Regression Models and Life-Tables. *Journal of the Royal Statistical Society*, 34(2), 187–220.
- [7] Kaplan, E. L., & Meier, P. (1958). Nonparametric Estimation from Incomplete Observations. *Journal of the American Statistical Association*, 53(282), 457–481.
- [8] Allison, P. D. (2010). *Survival Analysis Using SAS: A Practical Guide* (2nd ed.). SAS Institute.
- [9] Deloitte. (2025). *Annual review of football finance 2025*.  
<https://www.deloitte.com/content/dam/assets-zone2/uk/en/docs/services/consulting/2025/deloitte-annual-review-of-football-finance-2025.pdf>
- [10] Fédération Internationale de Football Association (FIFA). (2023). *2022 FIFA World Cup Qatar: Global engagement audience executive summary*.  
<https://digitalhub.fifa.com/m/336c30db79dafa93/original/FIFA-World-Cup-Qatar-2022-Global-Engagement-Audience-Executive-Summary.pdf>
- [11] Fynn, K. D., & Sonnenschein, M. (2012). An Analysis of the Career Length of Professional Basketball Players. 2.
- [12] Jones, M., Moteshare, A., Ball, S. V., Church, J. S., Calder, J. D. F., & Williams, A. (2025). Establishing 'normal' career longevity in professional footballers allows comparison to that of players with injuries and surgery. *Knee Surgery, Sports Traumatology, Arthroscopy*, 33(9), 3375–3388. <https://doi.org/10.1002/ksa.12722>
- [13] Kleinbaum, D. G., & Klein, M. (2012). Introduction to Survival Analysis. In D. G. Kleinbaum & M. Klein (Eds.), *Survival Analysis: A Self-Learning Text* (pp. 1–54, 97–145, 200–440). Springer.  
[https://doi.org/10.1007/978-1-4419-6646-9\\_1](https://doi.org/10.1007/978-1-4419-6646-9_1)
- [14] Staw, B. M., & Hoang, H. (1995). Sunk Costs in the NBA: Why Draft Order Affects Playing Time and Survival in Professional Basketball. *Administrative Science Quarterly*, 40(3), 474.  
<https://doi.org/10.2307/2393794>

[15] Ekstrand, J., Krutsch, W., Spreco, A., Van Zoest, W., Roberts, C., Meyer, T., & Bengtsson, H. (2020). Time before return to play for the most common injuries in professional football: A 16-year follow-up of the UEFA Elite Club Injury Study. *British Journal of Sports Medicine*, 54(7), 421–426.

[16] Webster, K. E., Hewett, T. E., & Feller, J. A. (2021). Return-to-play times and player performance after anterior cruciate ligament reconstruction in elite footballers: A systematic review. *Orthopaedic Journal of Sports Medicine*, 9(6), 1–11. (Open-access version: “Return-to-Play Times and Player Performance After ACL Reconstruction in Elite Soccer Players”)

[17] Hosmer, D. W., Lemeshow, S., & May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data* (2nd ed.). Wiley.

## Annex 1: EDA figures



